HARJOITUKSET – Tehtävä 4

Palauta word -dokumentti jossa kyselyt ja tulokset kuvakaappauksina tai terminaalista kopioituna Moodleen linkin kautta.

First be sure ae using right database –

write script *use sql_kyselyt;*

and then write *show tables;*

| *Tables_in_sql_kyselyt* |

+--------------------+

| *countries* |

| *departments* |

| *employees* |

| *job_history* |

| *jobs* |

| *locations* |

| *regions* |

+--------------------+

7 rows in set (0.00 sec)

So…we are ready to go challenges…

1. Write a SQL statement to create a simple table countries including columns country_id,country_name and region_id.

CREATE TABLE countries (COUNTRY_ID varchar(2) NOT NULL, COUNTRY_NAME varchar(40), REGION_ID decimal(10,0), PRIMARY KEY (COUNTRY_ID));

```
MariaDB [sql_kyselyt]> DESC countries;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| COUNTRY_ID  | varchar(2)   | NO   | PRI | NULL    |       |
| COUNTRY_NAME| varchar(40)  | YES  |     | NULL    |       |
| REGION_ID   | decimal(10,0)| YES  | MUL | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
3 rows in set (0.003 sec)
```

2. Write a SQL statement to create a simple table countries including columns country_id,country_name and region_id which is already exists.

CREATE TABLE IF NOT EXISTS countries (COUNTRY_ID varchar(2) NOT NULL, COUNTRY_NAME varchar(40), REGION_ID decimal(10,0), PRIMARY KEY (COUNTRY_ID));

```
MariaDB [sql_kyselyt]> CREATE TABLE IF NOT EXISTS countries ( COUNTRY_ID
 varchar(2) NOT NULL, COUNTRY_NAME varchar(40), REGION_ID decimal(10,0),
 PRIMARY KEY (COUNTRY_ID));
Query OK, 0 rows affected, 1 warning (0.000 sec)
```

3. Write a SQL statement to create the structure of a table dup_countries similar to countries.

```
MariaDB [sql_kyselyt]> CREATE TABLE dup_countries LIKE countries;
Query OK, 0 rows affected (0.018 sec)

MariaDB [sql_kyselyt]> SELECT * FROM dup_countries;
Empty set (0.006 sec)

MariaDB [sql_kyselyt]> DESC dup_countries;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| COUNTRY_ID  | varchar(2)   | NO   | PRI | NULL    |       |
| COUNTRY_NAME| varchar(40)  | YES  |     | NULL    |       |
| REGION_ID   | decimal(10,0)| YES  | MUL | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
3 rows in set (0.006 sec)
```

4. Write a SQL statement to create a duplicate copy of countries table including structure and data by name dup_countries.

```
MariaDB [sql_kyselyt]> INSERT INTO dup_countries SELECT * FROM countries;
Query OK, 25 rows affected (0.003 sec)
Records: 25  Duplicates: 0  Warnings: 0

MariaDB [sql_kyselyt]> SELECT * FROM dup_countries;
+------------+--------------------------+-----------+
| COUNTRY_ID | COUNTRY_NAME             | REGION_ID |
+------------+--------------------------+-----------+
| AR         | Argentina                |         2 |
| AU         | Australia                |         3 |
| BE         | Belgium                  |         1 |
| BR         | Brazil                   |         2 |
| CA         | Canada                   |         2 |
| CH         | Switzerland              |         1 |
| CN         | China                    |         3 |
| DE         | Germany                  |         1 |
| DK         | Denmark                  |         1 |
| EG         | Egypt                    |         4 |
| FR         | France                   |         1 |
| HK         | HongKong                 |         3 |
| IL         | Israel                   |         4 |
| IN         | India                    |         3 |
| IT         | Italy                    |         1 |
| JP         | Japan                    |         3 |
| KW         | Kuwait                   |         4 |
| MX         | Mexico                   |         2 |
| NG         | Nigeria                  |         4 |
| NL         | Netherlands              |         1 |
| SG         | Singapore                |         3 |
| UK         | United Kingdom           |         1 |
| US         | United States of America |         2 |
| ZM         | Zambia                   |         4 |
| ZW         | Zimbabwe                 |         4 |
+------------+--------------------------+-----------+
25 rows in set (0.000 sec)
```

5. Write a SQL statement to create a table countries set a constraint NULL.

CREATE TABLE IF NOT EXISTS countries (COUNTRY_ID varchar(2) NOT NULL, COUNTRY_NAME varchar(40) NULL , REGION_ID decimal(10,0), PRIMARY KEY (COUNTRY_ID));

6. Write a SQL statement to create a table named jobs including columns job_id, job_title, min_salary, max_salary and check whether the max_salary amount exceeding the upper limit 25000.

CREATE TABLE IF NOT EXISTS `jobs` (JOB_ID varchar(10) NOT NULL DEFAULT '', JOB_TITLE varchar(35) NOT NULL, MIN_SALARY decimal(6,0) DEFAULT NULL, MAX_SALARY decimal(6,0) DEFAULT NULL, PRIMARY KEY (`JOB_ID`));

```
MariaDB [sql_kyselyt]> SELECT * FROM jobs WHERE max_salary > 25000;
+---------+-------------------------------+------------+------------+
| JOB_ID  | JOB_TITLE                     | MIN_SALARY | MAX_SALARY |
+---------+-------------------------------+------------+------------+
| AD_PRES | President                     |      20000 |      40000 |
| AD_VP   | Administration Vice President |      15000 |      30000 |
+---------+-------------------------------+------------+------------+
2 rows in set (0.007 sec)
```

SELECT statement

7. Write a query to display the names (first_name, last_name) using alias name “First Name", "Last Name" Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT first_name AS "First Name", last_name AS "Last Name" FROM employees LIMIT 5;
+------------+-----------+
| First Name | Last Name |
+------------+-----------+
| Ellen      | Abel      |
| Sundar     | Ande      |
| Mozhe      | Atkinson  |
| David      | Austin    |
| Hermann    | Baer      |
+------------+-----------+
5 rows in set (0.001 sec)
```

Put limit cause list is so long.

8. Write a query to get unique department ID from employee table. Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT department_id FROM employees LIMIT 8;
+---------------+
| department_id |
+---------------+
|             0 |
|            10 |
|            20 |
|            20 |
|            30 |
|            30 |
|            30 |
|            30 |
+---------------+
8 rows in set (0.000 sec)
```

Put limit cause list is so long.

9. Write a query to get all employee details from the employee table order by first name, descending. Sample table: employees

SELECT * FROM employees ORDER BY first_name DESC;

10. Write a query to get the names (first_name, last_name), salary, PF of all the employees (PF is calculated as 12% of salary). Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT first_name, last_name, salary, salary*0.12 AS PF FROM employees LIMIT 10;
+------------+-----------+----------+-----------+
| first_name | last_name | salary   | PF        |
+------------+-----------+----------+-----------+
| Steven     | King      | 24000.00 | 2880.0000 |
| Neena      | Kochhar   | 17000.00 | 2040.0000 |
| Lex        | De Haan   | 17000.00 | 2040.0000 |
| Alexander  | Hunold    |  9000.00 | 1080.0000 |
| Bruce      | Ernst     |  6000.00 |  720.0000 |
| David      | Austin    |  4800.00 |  576.0000 |
| Valli      | Pataballa |  4800.00 |  576.0000 |
| Diana      | Lorentz   |  4200.00 |  504.0000 |
| Nancy      | Greenberg | 12000.00 | 1440.0000 |
| Daniel     | Faviet    |  9000.00 | 1080.0000 |
+------------+-----------+----------+-----------+
10 rows in set (0.000 sec)
```

Put limit cause list is so long.

11. Write a query to get the employee ID, names (first_name, last_name), salary in ascending order of salary. Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT employee_id, CONCAT(first_name,' ' ,last_name) AS names, salary FROM employees ORDER BY salary ASC LIMIT 10;
+-------------+-----------------+--------+
| employee_id | names           | salary |
+-------------+-----------------+--------+
|         132 | TJ Olson        | 2100.00 |
|         128 | Steven Markle   | 2200.00 |
|         136 | Hazel Philtanker | 2200.00 |
|         127 | James Landry    | 2400.00 |
|         135 | Ki Gee          | 2400.00 |
|         144 | Peter Vargas    | 2500.00 |
|         119 | Karen Colmenares | 2500.00 |
|         131 | James Marlow    | 2500.00 |
|         140 | Joshua Patel    | 2500.00 |
|         182 | Martha Sullivan | 2500.00 |
+-------------+-----------------+--------+
10 rows in set (0.001 sec)
```

Put limit cause list is so long.

12. Write a query to get the total salaries payable to employees.Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT SUM(salary) AS total_salaries FROM employees;
+----------------+
| total_salaries |
+----------------+
|      691400.00 |
+----------------+
1 row in set (0.001 sec)
```

13. Write a query to get the maximum and minimum salary from employees table. Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT MAX(salary) AS maximum_salary, MIN(salary) AS minimum_salary FROM employees;
+---------------+---------------+
| maximum_salary | minimum_salary |
+---------------+---------------+
|       24000.00 |        2100.00 |
+---------------+---------------+
1 row in set (0.002 sec)
```

14. Write a query to get the average salary and number of employees in the employees table. Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT AVG(salary) AS average_salary,COUNT(employee_id) AS "number of employees" FROM employees;
+----------------+---------------------+
| average_salary | number of employees |
+----------------+---------------------+
|    6461.682243 |                 107 |
+----------------+---------------------+
1 row in set (0.002 sec)
```

15. Write a query to get the number of employees working with the company. Sample table: employees

```
MariaDB [sql_kyselyt]> SELECT COUNT(employee_id) AS "number of employees working" FROM employees WHERE salary > 1000;
+----------------------------+
| number of employees working |
+----------------------------+
|                        107 |
+----------------------------+
1 row in set (0.001 sec)
```

16. Write a query to get the number of jobs available in the employees table.

```
MariaDB [sql_kyselyt]> SELECT COUNT(DISTINCT(job_id)) AS "number of jobs" FROM employees;
+-----------------+
| number of jobs |
+-----------------+
|              19 |
+-----------------+
1 row in set (0.007 sec)
```

17. Write a query to get monthly salary (round 2 decimal places) of each and every employee. Note : Assume the salary field provides the 'annual salary' information

```
MariaDB [sql_kyselyt]> SELECT employee_id, ROUND(salary,2) As salary from employees LIMIT 5;
+-------------+----------+
| employee_id | salary   |
+-------------+----------+
|         100 | 24000.00 |
|         101 | 17000.00 |
|         102 | 17000.00 |
|         103 |  9000.00 |
|         104 |  6000.00 |
+-------------+----------+
5 rows in set (0.000 sec)
```

18. Write a SQL statement to change salary of employee to 8000 whose ID is 105, if the existing salary is less than 5000. Sample table: employees **. Hint. Use Update!**

```
MariaDB [sql_kyselyt]> UPDATE employees SET salary = if(salary<5000,8000,salary) WHERE employee_id = 105;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [sql_kyselyt]> UPDATE employees SET salary = if(salary<5000,8000) WHERE employee_id = 105;
MariaDB [sql_kyselyt]> select * from employees where employee_id = 105;
+-------------+------------+-----------+---------+--------------+------------+---------+---------+----------------+------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL   | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY  | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------+--------------+------------+---------+---------+----------------+------------+---------------+
|         105 | David      | Austin    | DAUSTIN | 590.423.4569 | 1987-06-22 | IT_PROG | 8000.00 |           0.00 |        103 |            60 |
+-------------+------------+-----------+---------+--------------+------------+---------+---------+----------------+------------+---------------+
1 row in set (0.001 sec)
```